

NUS Coding Workshop: Copilot

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National University of Singapore

About me

- Shao Yurui
- Electrical & Computer Engineering
- Freshly graduated from NUS last year 2024
- Full-time Teaching Assistant
- Ph.D. Candidate in ML for healthcare



Fun Fact about Me #1

- My parents are living in the northern part of China with a super cute Shiba Inu



Fun Fact about Me #2



- I worked as a Bartender during Year 1 December Break (circuit breaker)



Content Table

- 1. Why use AI**
- 2. History of Large Language Model (LLM)**
- 3. Coding Agent -- VS Code Copilot**
- 4. NUS AI Policy**
- 5. Roadmap of Learning AI**
- 6. Hands-on Activity**



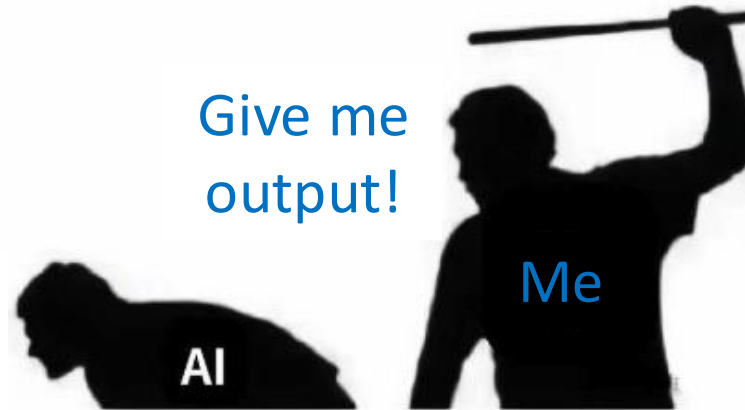
Why use AI

5 mins

Why learn to use AI

What you imagine life will be ..

I want to exterminate
humanity.



And the reality:

AI can do all of this.
So what exactly are you here for?

Sorry boss...



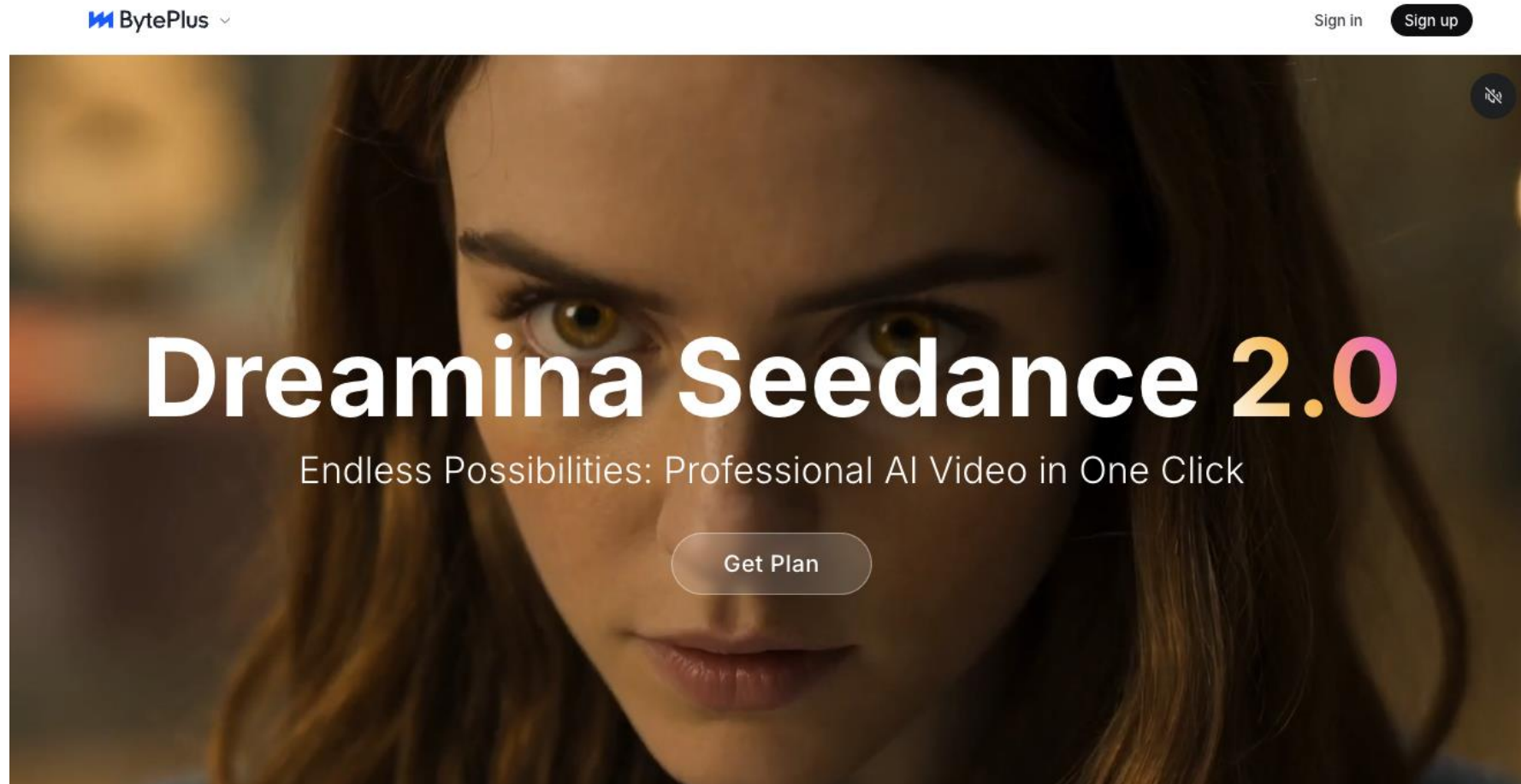


Demo of latest AI Tools

5 mins

Demo1: Seedance, Your film director

<https://www.byteplus.com/en/activity/seedance2-0>



Demo1: StableDiffusion (2023)



Demo1: Seedance 2.0 (2026)



Demo1: Seedance 2.5 (Aug 2026)



Demo2: NotebookLM, your personal tutor

<https://notebooklm.google/>

 Gemini Notebook



Understand **anything**

Your research and thinking partner, grounded in the information that you trust, built with the latest Gemini models.

Try Gemini Notebook

Demo 3: Agent, your personal assistant

The screenshot displays the Claude AI interface within a web browser. The left sidebar shows a project list with 'LLM development history' selected. The main chat area contains a message from the user and a response from Claude. A document titled 'History of LLM Development' is open on the right, showing a table of milestones.

Workshop o... / LLM development ...

Home Code

+ New

Projects

Artifacts

Scheduled

Dispatch Beta

Customize

Projects

Workshop on AI for school work

AI workshop structure

LLM development history

Recents >

Design

Rio Pro

now lets first work on the section of hisotry of LLM development. along the way I want to explain keywords like "LLM, transformers, token, API, agent, skill"

Read a memory, used a tool

What format should this History of LLM Development section be delivered in? Speaker/handout doc (.docx or Markdown)

How deep should the audience's technical background be assumed given they're university students in a general 'AI for school

Write a message...

+ Sonnet 5 High

Claude is AI and can make mistakes. Please double-check responses. [Give us feedback](#)

History of llm d... Download and open

SECTION 1

History of LLM Development

A 10-minute walkthrough, with six keywords explained along the way: LLM, Transformer, Token, API, Agent, Skill.

Why This History Matters

Students often meet AI tools as a grab-bag of unrelated apps — ChatGPT here, NotebookLM there, Copilot somewhere else. In reality, almost everything on the market today is built on the same handful of ideas, developed over roughly the last decade. Once you understand where these tools came from, you can reason about any new AI product you meet, instead of re-learning from scratch each time. This section moves fast — the goal is recognition, not a deep technical dive.

The Timeline (10-Minute Walkthrough)

Read across each row left to right. The right-hand column flags the keyword to introduce and define as you reach that row (full definitions are in the glossary below).

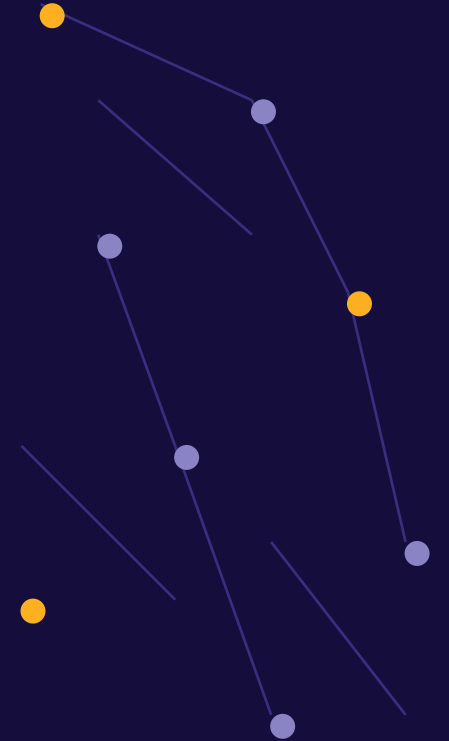
Year	Milestone	Talking point (say this out loud)	Keyword
1960s	ELIZA, the first chatbot	Simple pattern-matching, not real understanding — it just reflected your words back as questions. Sets the baseline: for decades, computers could not genuinely handle language.	
2017	The Transformer is invented	Google researchers publish "Attention Is All You Need." This new architecture lets a model weigh how every word relates to every other word at once, instead of reading strictly left-to-right. It is the engine under almost every AI tool you use today.	Transformer
2018–2020	GPT-1 → GPT-2 → GPT-3	Same basic idea, scaled up massively — more text, more computing power. GPT-3 (2020) was the first model fluent enough to feel genuinely useful, though it still lived behind the scenes in research labs and select apps.	LLM, Token
Nov 2022	ChatGPT launches	OpenAI puts a chat window in front of GPT-3.5. It hits 100 million users within two months — the moment AI went from "research curiosity" to something students, teachers, and offices were using daily.	
2023	GPT-4 and Claude arrive; apps plug in	Models get sharper and multimodal (they can read images, not just text). Just as important: companies open up API access, so any app — a study tool, a browser extension, a note-taking app — can call the model behind the scenes.	API
2024	Reasoning and multimodal models	Models like OpenAI's o1 learn to "think before answering" on hard problems (math, code, logic), and everyday models get much better at images, audio, and long documents.	

Page 1 / 3

SECTION 1

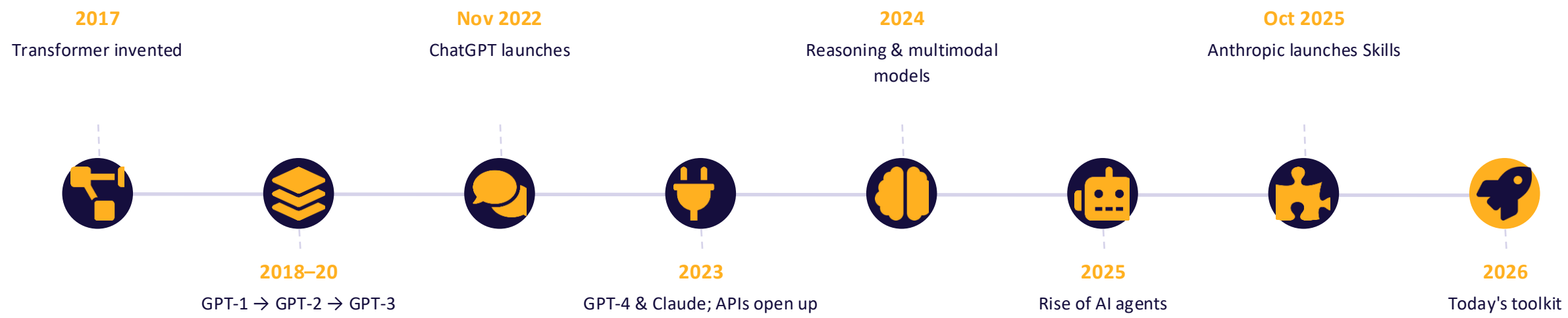
History of LLM Development

From a 2017 research paper to the AI tools you use every day



The Road Here — At a Glance

Eight moments, five years. We'll take each one in turn.





2017

MILESTONE 1 OF 8

The Transformer Is Invented

Google researchers publish “Attention Is All You Need.” The idea: let a model weigh how every word in a passage relates to every other word, all at once — instead of reading strictly left to right. It becomes the engine under almost every AI tool you use today.

Take the sentence: "The cat sat on the mat because it was tired." When you read the word "it", your brain instantly connects it back to "the cat", not "the mat".

“

Attention Is All You Need

— Vaswani et al., Google, 2017

KEYWORD: TRANSFORMER



**2018–2020**

MILESTONE 2 OF 8

GPT-1 → GPT-2 → GPT-3

Same basic idea, scaled up massively — more text, more computing power.
GPT-3 (2020) was the first model fluent enough to feel genuinely useful,
though it still lived behind the scenes in research labs and select apps.

Parameters (scale, not to scale)

GPT-1

117M params

GPT-2

1.5B params

GPT-3

175B params

KEYWORD: LLM

KEYWORD: TOKEN



Nov 2022

MILESTONE 3 OF 8

ChatGPT Goes Mainstream

OpenAI puts a chat window in front of GPT-3.5. It hits 100 million users within two months — the moment AI went from a research curiosity to something students, teachers, and offices were using daily.

100M

users in the first two months

Launched Nov 30, 2022



2023

MILESTONE 4 OF 8

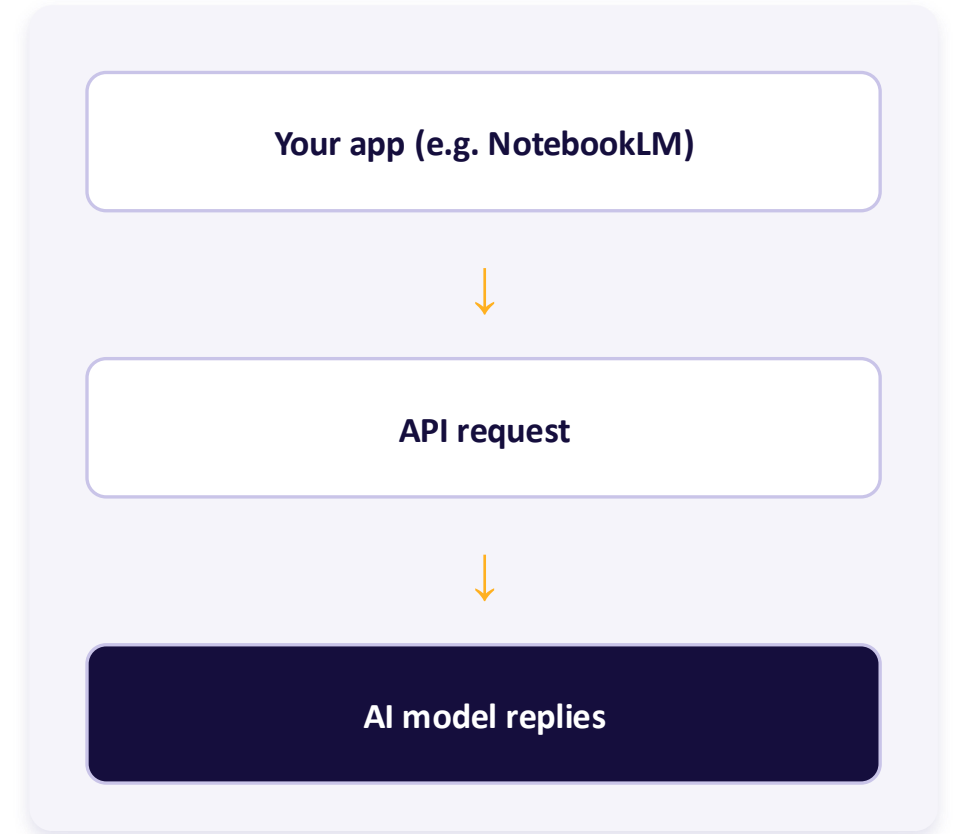
History of LLM Development

GPT-4 & Claude Arrive — Apps Plug In

Models get sharper and multimodal (they can read images, not just text). Just as important: companies open up API access, so any app — a study tool, a browser extension, a note-taking app — can call the model behind the scenes.

Our new keyword here is API, which stands for Application Programming Interface.

KEYWORD: API





2024

MILESTONE 5 OF 8

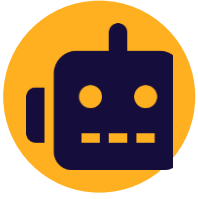
Reasoning & Multimodal Models

Models like OpenAI's o1 learn to “think before answering” on hard problems — math, code, logic. Meanwhile everyday models get much better at images, audio, and long documents in a single conversation.

“

Now it can see, hear, and think step by step — not just autocomplete.

— the shift from 2023 to 2024 models



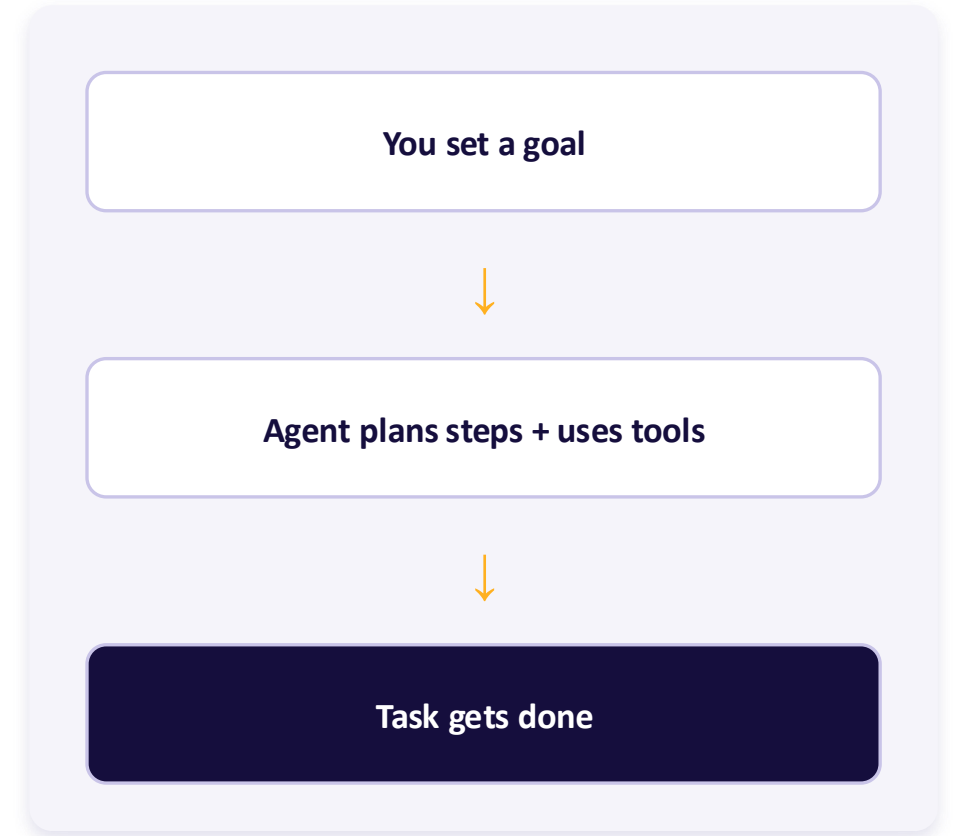
2025

MILESTONE 6 OF 8

The Rise of AI Agents

Models stop just answering questions and start doing multi-step work: browsing, running code, editing files, checking their own results — with a person still setting the goal and staying in the loop.

KEYWORD: AGENT





Oct 2025

MILESTONE 7 OF 8

Anthropic Launches Agent Skills

Instead of re-explaining a task every time, you can package instructions once as a reusable “Skill” that an agent pulls out whenever that kind of task comes up — like a recipe card for a familiar job.

KEYWORD: SKILL

SKILL.md

name:

apa-citation-formatter

description:

Format citations in APA 7th ed. Use when a user asks to format references or check APA style.

Only this ~100-token header loads until it's needed.





2026

MILESTONE 8 OF 8

Where We Are Now

A crowded, fast-moving toolkit — Claude, ChatGPT, Gemini, GitHub Copilot, and more — all built on the ideas in this timeline. That's exactly what we'll tour next.

Today's toolkit

Claude

ChatGPT

Gemini / Gemini Notebook

GitHub Copilot

Six Words, One Story

Quick recap before we move on

LLM

A program trained on huge amounts of text to predict and generate language.

Transformer

The design that lets a model weigh how every word relates to every other, at once.

Token

The small text chunks a model reads and writes — the Lego bricks of language.

API

How one app hands your request to an AI model and gets a reply back.

Agent

An AI that takes a goal, plans steps, and uses tools to get a task done.

Skill

A packaged, reusable set of instructions an agent loads when a task calls for it.



VS Code Copilot

10 mins

Copilot: Your coding agent

The screenshot displays the Visual Studio Code interface with three main components:

- Files (Left Panel):** Shows a file explorer with a project structure including folders like `.vscode`, `data`, and `src`. The `src` folder contains various files such as `augmentation.ipynb`, `buzzer_demo.ipynb`, `data_exploration.ipynb`, `DataVisualisation.ipynb`, `fols_preprocess_comp...`, `Impact_of_Preprocessi...`, `Inn_split_test.ipynb`, `Inn_testing-Copy1.ipynb`, `Inn_testing.ipynb`, `LNN_Window_target_I...`, `LNN_Window_targetip...`, `main.ipynb`, `main.py`, `model_test.ipynb`, `my_functions.ipynb`, `notebook.ipynb`, `pipeline.ipynb`, `preproc111ss.ipynb`, `preprocess.ipynb`, `rnn_model.pth`, `S121_trial_1_S_L.csv`, `scanner.ipynb`, and `Serial_com.ipynb`.
- Python Notebook (Center Panel):** Displays the `DataVisualisation.ipynb` notebook. It shows three code cells:
 - Cell [7]: A function `readDataBase()` that iterates through a directory, reads CSV files, and appends them to a list of dataframes.
 - Cell [8]: A call to `my_dataframes = readDataBase()`.
 - Cell [36]: A plot of `my_dataframes[0]['Gyro_x_left']` using `plt.figure(figsize=(12, 4))`, `plt.plot()`, `plt.title()`, `plt.xlabel()`, `plt.ylabel()`, and `plt.show()`.The plot shows a time series of gyro data with a title "Gyro_x_left from my_dataframes[0]".
- Copilot (Right Panel):** Shows the Copilot chat interface. It displays a chat history with a message: "Similar code found with 1 license type - View matches". Below this, it shows a code snippet: `@workspace plot out my_dataframes[0]['Gyro_x_left']`. The chat also includes a section titled "Workspace" with a message: "You can plot my_dataframes[0]['Gyro_x_left'] using matplotlib as follows:" followed by a code snippet: `# ...existing code... plt.figure(figsize=(12, 4)) plt.plot(my_dataframes[0]['Gyro_x_left']) plt.title('Gyro_x_left from my_dataframes[0]') plt.xlabel('Index') plt.ylabel('Gyro_x_left') plt.show() # ...existing code...`. At the bottom, there is a section titled "Ask Copilot" with a dropdown menu set to "GPT-4.1".

Files

Python Notebook

Copilot

Install Visual Studio Code (VScode)

- <https://code.visualstudio.com/Download> (Just google VScode)



Visual Studio Code

Docs

Updates

Blog

API

Extensions

FAQ

GitHub Copilot

MCP



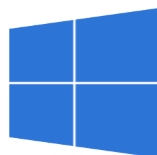
Search Docs

Download

Try [MCP servers](#) to extend agent mode in VS Code!

Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



↓ Windows

Windows 10, 11

User Installer [x64](#) [Arm64](#)
System
Installer [x64](#) [Arm64](#)
.zip [x64](#) [Arm64](#)



↓ .deb

Debian, Ubuntu

↓ .rpm

Red Hat, Fedora, SUSE

.deb [x64](#) [Arm32](#) [Arm64](#)
.rpm [x64](#) [Arm32](#) [Arm64](#)
.tar.gz [x64](#) [Arm32](#) [Arm64](#)
Snap [Snap Store](#)



↓ Mac

macOS 11.0+

.zip [Intel chip](#) [Apple silicon](#) [Universal](#)
CLI [Intel chip](#) [Apple silicon](#)

Install Anaconda

- <https://www.anaconda.com/download> (Just google Anaconda)



Distribution

Register to get everything you need to get started on your workstation including Cloud Notebooks, Navigator, AI Assistant, Learning and more.

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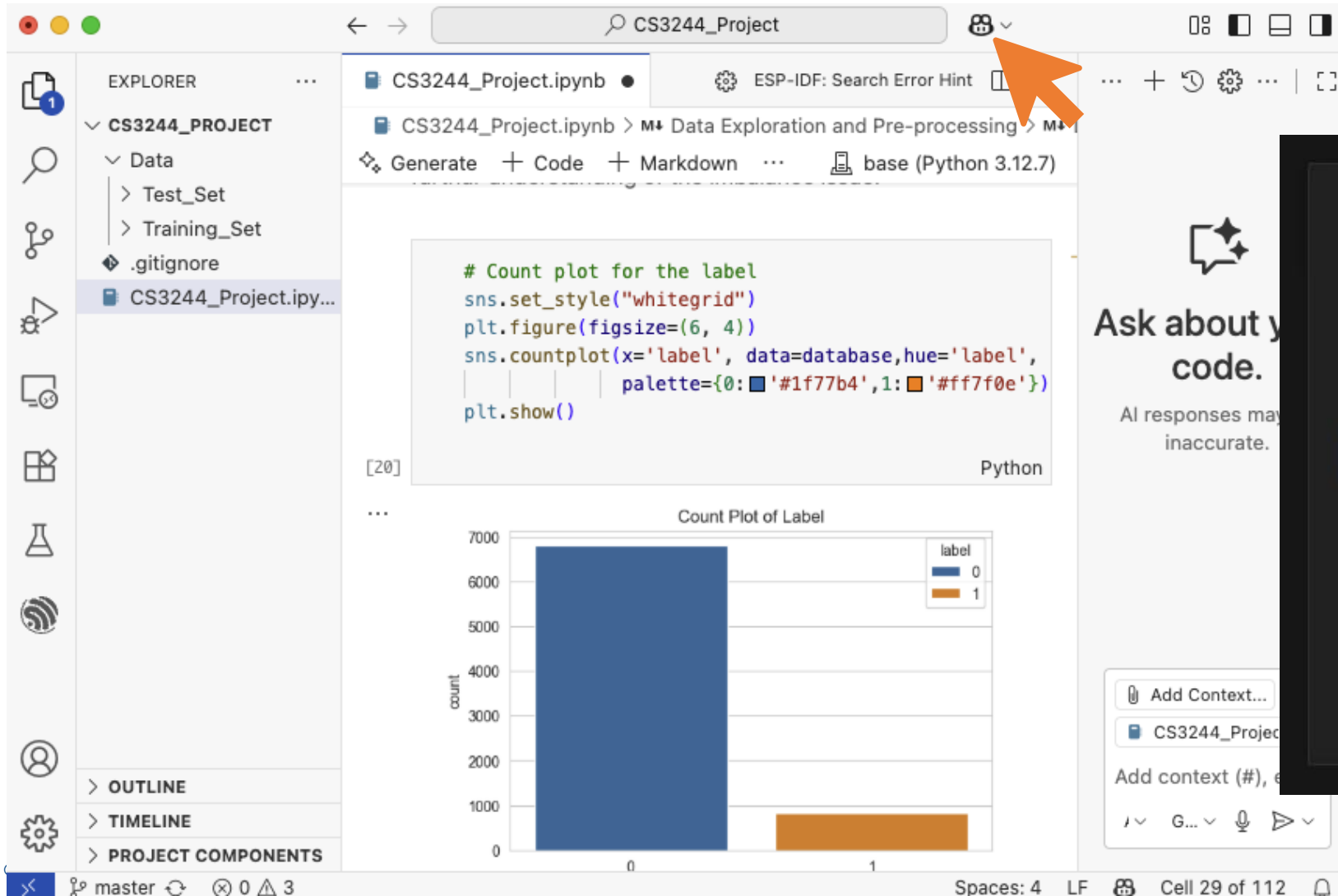
Returning Users >

*Subject to our [Terms of Service](#). Use of Anaconda's offerings at an organization of more than 200 employees/contractors requires a paid business license unless your organization is eligible for discounted or free use. [See Pricing](#).

[Skip registration](#)



Sign up GitHub Account



The screenshot shows a Jupyter Notebook interface with a file explorer on the left, a code editor in the center, and a sidebar on the right. The file explorer shows a project named 'CS3244_PROJECT' with subfolders 'Data' (containing 'Test_Set' and 'Training_Set') and a file 'CS3244_Project.ipynb'. The code editor displays a Python script for a count plot. The sidebar on the right shows a GitHub Copilot overlay with a 'Continue with GitHub' button.

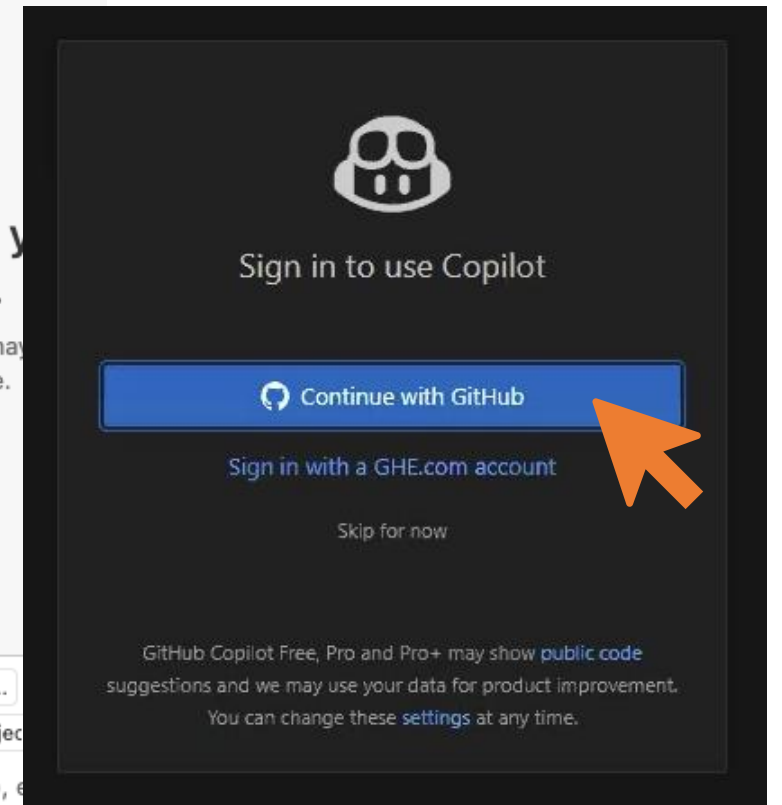
```
# Count plot for the label
sns.set_style("whitegrid")
plt.figure(figsize=(6, 4))
sns.countplot(x='label', data=database, hue='label',
              palette={0: '#1f77b4', 1: '#ff7f0e'})
plt.show()
```

[20] Python

Count Plot of Label

label	count
0	6800
1	800

Spaces: 4 LF Cell 29 of 112



Sign in to use Copilot

[Continue with GitHub](#)

[Sign in with a GHE.com account](#)

[Skip for now](#)

GitHub Copilot Free, Pro and Pro+ may show [public code](#) suggestions and we may use your data for product improvement. You can change these [settings](#) at any time.

Sign up GitHub Account

- <https://github.com/signup> (Just google GitHub)

Create your free account

Explore GitHub's core features for individuals and organizations.

See what's included ▾



Sign up to GitHub

Email*

Email

Use school email !!

Password*

.....

Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Username*

ShaoYurui

Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.

Your Country/Region*

Vietnam ▾

For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

Email preferences

☐ Receive occasional product updates and announcements

Create account >



GitHub Education Benefit with NUS email

 **Yurui (ShaoYurui)**
Your personal account

[Go to your personal profile](#)

-  Public profile
-  Account
-  Appearance
-  Accessibility
-  Notifications

Access

 **Billing and licensing** ^

- Overview
- Usage
- Budgets and alerts
- Licensing
- Payment information
- Payment history
- Additional billing details
- Education benefits**
-  Emails
-  Password and authentication
-  Sessions
-  SSH and GPG keys

GitHub Education



Education Benefits

You have a current student coupon applied. Find more information on your benefits [here!](#)

[Start an application](#)



Coupon applied



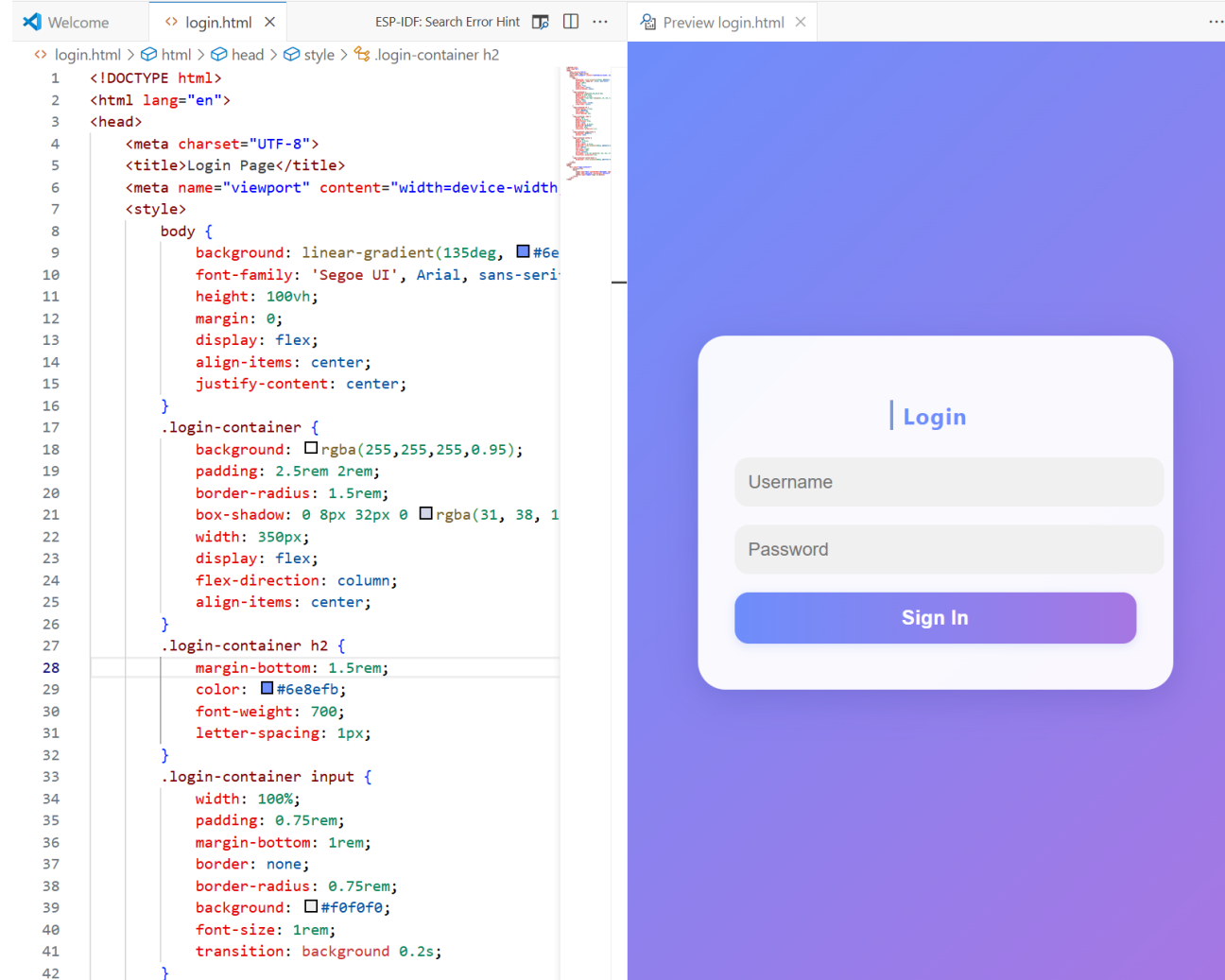
Expires in almost 2 years



Features of Copilot

15 mins

Generate Code (ASK)



```
<?xml version="1.0" encoding="UTF-8" ?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Login Page</title>
<meta name="viewport" content="width=device-width, height=device-height">
<style>
body {
background: linear-gradient(135deg, #6e8efb, #a6c1ee);
font-family: 'Segoe UI', Arial, sans-serif;
height: 100vh;
margin: 0;
display: flex;
align-items: center;
justify-content: center;
}
.login-container {
background: #fff;
padding: 2.5rem 2rem;
border-radius: 1.5rem;
box-shadow: 0 8px 32px 0 #313841, 0 0 0 1px #313841;
width: 350px;
display: flex;
flex-direction: column;
align-items: center;
}
.login-container h2 {
margin-bottom: 1.5rem;
color: #6e8efb;
font-weight: 700;
letter-spacing: 1px;
}
.login-container input {
width: 100%;
padding: 0.75rem;
margin-bottom: 1rem;
border: none;
border-radius: 0.75rem;
background: #f0f0f0;
font-size: 1rem;
transition: background 0.2s;
}
.login-container button {
width: 100%;
padding: 0.75rem;
margin-top: 1rem;
border: none;
border-radius: 0.75rem;
background: #4a7ebb;
color: white;
font-weight: 700;
font-size: 1rem;
}

```

CHAT

create a html file, showing a login page with user name and password, use fancy styling

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Login Page</title>
  <meta name="viewport" content="width=device-width, height=device-height">
  <style>
    body {
      background: linear-gradient(135deg, #6e8efb, #a6c1ee);
      font-family: 'Segoe UI', Arial, sans-serif;
      height: 100vh;
      margin: 0;
      display: flex;
      align-items: center;
    }
  
```

Generate Code (AGENT)

The screenshot shows a code editor with the following HTML and CSS code:

```
<> html > html > head > style > button
1 <!-- You can save this as login.html and open in your browser -->
2 <!DOCTYPE html>
3 <html lang="en">
4 <head>
5   <meta charset="UTF-8">
6   <title>Login Page</title>
7   <link href="https://fonts.googleapis.com/css?family=Montserrat:"
8   <style>
9     body { font-family: Arial, sans-serif; background:
10    body {
11      font-family: 'Montserrat', Arial, sans-serif;
12      background: linear-gradient(135deg, #74ebd5 0%, #AC
13      min-height: 100vh;
14      margin: 0;
15      display: flex;
16      align-items: center;
17      justify-content: center;
18    }
19    .login-container {
20      width: 300px;
21      margin: 100px auto;
22      background: #fff;
23      padding: 30px;
24      border-radius: 8px;
25      box-shadow: 0 0 10px rgba(0,0,0,0.1);
26    }
27    width: 350px;
28    background: rgba(255,255,255,0.95);
29    padding: 40px 30px 30px 30px;
30    border-radius: 18px;
```

The right-hand panel shows a preview of the login page with the text "use fancy styling". Below the preview, there is a chat interface with the following text:

<> html +69 -16

The login page now uses fancy styling with gradients, rounded corners, modern input effects, and a stylish button. Open the file in your browser to see the enhanced design. If you need further customization or features, let me know!

1 file changed Keep Undo

<> html +

Add Context (#), extensions (@), commands (/)

Agent GPT-4.1

Col 25 Spaces: 4 UTF-8 {} HTML

Fix Bugs

```
Welcome | login.html | fix demo.py [icon] X
fix demo.py > ...
1  import tkinter as tk
2
3  # Create the main window
4  root = tk.Tk()
5  root.title("Python Graphic Demo")
6  root.geometry("300x200")
7
8  # Add a label
9  label = tk.Label(root, text="Hello, World!", font=("Segoe UI",
10 label.pack(pady=40)
11
12 # Add a button
13 def on_click():
14     label.config(text="Button Clicked!")
15
16 button = tk.Button(root, text="Click Me", commaand=on_click, fo
17 button = tk.Button(root, text="Click Me", command=on_click, for
18 button.pack(pady=10)
19
20 # Start the main event loop
21 root.mainloop()
```

@workspace /fix

fix demo.py:1-20

Workspace

> Used 1 reference

Pseudocode Plan:

1. Import tkinter.
2. Create main window, set title and geometry.
3. Add a label.
4. Define button click handler.
5. Add a button (fix typo in 'command' argument).

Code:

```
import tkinter as tk

# Create the main window
root = tk.Tk()
```



“Context”: What is known to Copilot

```
demo_workspace.py U x ESP-IDF: Search Error Hint ...  
demo_workspace.py  
1 from hello import print_hello  
2  
3 print_hello()
```

```
/demo_workspace.py  
ASJDBIyf98a7s6dyriugasdf7i843
```

```
hello.py U x ...  
hello.py > print_hello  
1 def print_hello():  
2     print("ASJDBIyf98a7s6dyriugasdf7i843")
```

@workspace /fix print hello()
demo_workspace.py

</> Workspace
> Used 1 reference

Context (#)

demo_workspace.py U × ESP-IDF: Search Error Hint

demo_workspace.py

```
1 from hello import print_hello
2
3 print_hello()
```

CHAT

#file:demo_workspace.py #file:demo_fix.py fix print hello

demo_fix.py

demo_workspace.py

demo_workspace.py

> Used 2 references

To fix the `print_hello` function, you need to ensure that the `hello.py` module exists and defines a `print_hello` function. Here's how you can create or update `hello.py`:

```
def print_hello():
    print("Hello, World!")
```

Now, your `demo_workspace.py` will work as expected:

Add Context...

demo_workspace.py ×

Add context (#), extensions (@), commands (/)

Ask GPT-4.1



Context (highlight)

The image shows a VS Code editor window with a Python file named `demo_fix.py`. The code is a Tkinter application that creates a window, adds a label, a button, and starts the main event loop. A red rectangle highlights lines 12-14, which define the `on_click` function. To the right, a chat window is open, showing the same code snippet. A red rectangle in the chat window highlights the context selection dropdown, which shows `demo_fix.py:12-14`.

```
demo_fix.py > ...
1  import tkinter as tk
2
3  # Create the main window
4  root = tk.Tk()
5  root.title("Python Graphic Demo")
6  root.geometry("300x200")
7
8  # Add a label
9  label = tk.Label(root, text="Hello, World!", font=("Segoe
10 label.pack(pady=40)
11
12 # Add a button
13 def on_click():
14     label.config(text="Button Clicked!")
15
16 button = tk.Button(root, text="Click Me", command=on_click)
17 button.pack(pady=10)
18
19 # Start the main event loop
20 root.mainloop()
```

CHAT

```
root.geometry("300x200")

# Add a label
label = tk.Label(root, text="Hello, World
label.pack(pady=40)

# Add a button
def on_click():
    label.config(text="Button Clicked!")

button = tk.Button(root, text="Click Me",
button.pack(pady=10)

# Start the main event loop
root.mainloop()
```

Add Context ... demo_fix.py:12-14

Add context (#), extensions (@), commands (/)

Ask GPT-4.1

Context

"Dear Professors,

Good day, hope you are doing well ! I would like to inform you that I will be on leave from 18 Jul to 31 Jul. Thank you so much for the continued support!

Thank you for your time and support.

Sincerely

Yurui"

This is the context you provided

Improve this email |



Context

Introducing GPT-5

ChatGPT now has our smartest, fastest, most useful model yet, with thinking built in — so you get the best answer, every time.

```
"# Add a button  
def on_click():  
    label.config(text="Button Clicked!")"
```

This is the context you provided

Explain this code



Learning with Copilot

demo.py > ...

```
1  # Simple Flask server for login
2  from flask import Flask, render_template_string, request
3
4  app = Flask(__name__)
5
6  LOGIN_HTML = """
7  <!DOCTYPE html>
8  <html>
9  <body>
10     <form method="POST">
11         <input name="username" placeholder="Username" required>
12         <input name="password" type="password" placeholder="Password" required>
13         <button type="submit">Login</button>
14     </form>
15     {% if message %}
16     <p>{{ message }}</p>
17     {% endif %}
18 </body>
19 </html>
20 """
21
22 USERS = {"user1": "pass1", "user2": "pass2"}
23
24 @app.route("/", methods=["GET", "POST"])
25 def login():
26     message = ""
27     if request.method == "POST":
```

I want to learn about how to set up a server and handle basic login and password, give me some exercises and example to learn

<> login.html:1-42

> Used 1 reference

Here are some beginner exercises and examples to help you learn about setting up a server and handling basic login/password:

Exercises

1. Set up a simple HTTP server

- Use Python's `http.server` or Node.js `express` to serve your HTML login page.

2. Handle form submission

- Modify your login page to POST username and password to the server.
- On the server, print the received data.



Summary

- **Mode**
 - Ask
 - Agent
- **Slash Command**
 - \fix
 - \tests
- **Context**
 - Cursor Highlight
 - # Files
 - @workspace



Summary

Additional Resources

- Get Started with GitHub Copilot in VS Code (2025)
<https://www.youtube.com/watch?v=vdBxfFVXnc0&t=625s>
- Copilot Best Practices (What Not To Do)
<https://www.youtube.com/watch?v=2q0BoioYSxQ>



NUS AI Policy

5 mins



NUS AI Policy

- Policy for Use of AI in Teaching & Learning (2026)

<https://ctlit.nus.edu.sg/ai-community-of-practice/policies/>

- The NUS AI Guidelines for Students can be found as follows:

https://libguides.nus.edu.sg/new2nus/ai_guidelines_infographics



NUS AI Policy

- Guidelines on the Use of AI Tools for Academic Work
 - **Don't Use AI to Plagiarize**
 - Generating an output and presenting it as your own work or idea without attribution.
 - **Proper uses of AI tools**
 - Gathering information and looking up explanations for basic concepts.
 - Generating output for critique and analysis, for self-learning, or to compare against one's work for self-evaluation and improvement.
 - Help with reading and editing writing
 - **Acknowledging your use of AI**

Acknowledging your use of AI

You can give this acknowledgement through a note or “methods section” at the end of the assignment explaining, e.g., which AI tools were used, in which parts of the process they were used, what were the prompts used to generate results, and what you did with the outputs to add value. One way this can be done is in a tabular form as shown below:

AI Tool Used	Prompt and output	How the output is used in the assignment

- Policy for Use of AI in Teaching & Learning (2024)
<https://ctlit.nus.edu.sg/ai-community-of-practice/policies/>
- Page 7

Example: Acknowledging your use of AI

AI Tool Used	Prompt and output	How the output is used in the assignment
ChatGPT	Input: read csv in python output: In Python, the most common way to read a CSV file is with pandas. Here's the basic example: import pandas as pd # Read a CSV file df = pd.read_csv("file.csv") # Display the first 5 rows print(df.head())	Paste and modify to read csv file

NUS AI Policy:

**Never dump the course materials
into ChatGPT (notes, tutorials ...)**

ChatGPT ▾

What are you working on?

+ solve this:|



EE2026_Tutorial-
5 (wk7)_2024

NUS AI Policy: Free ChatGPT

AI in Education



Summarise



✉ Provost's Circular Manager <provostscircular@nus.edu.sg>

Tuesday, 11 August 2026 at 10:57

To support this, we are taking a portfolio approach to AI, where a range of tools will be available. We have signed an agreement with OpenAI to provide **ChatGPT** Edu access to you. Students in courses requiring advanced tools such as Codex and Rosalind will also be provided with credits to use these tools.

Details on access, together with other information and support, will be provided on **31 August 2026**.

Thank you.

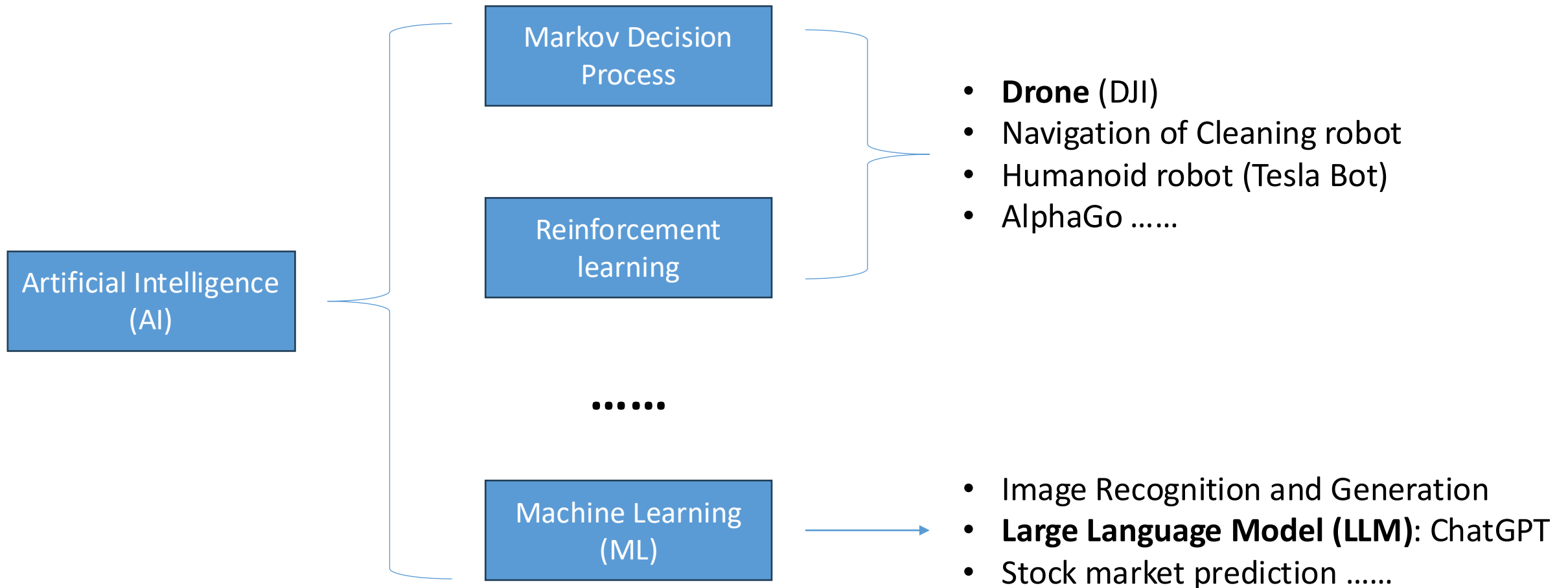
Sincerely,
Professor Aaron Thean
Deputy President (Academic Affairs) and Provost



Roadmap of Learning AI

5 mins

Overview of AI



Development of ML

Machine Learning
(ML)

CS2102
Database Systems

1960s

Classical
Machine Learning

1967

Neural Network
Multilayer Perceptron

1989

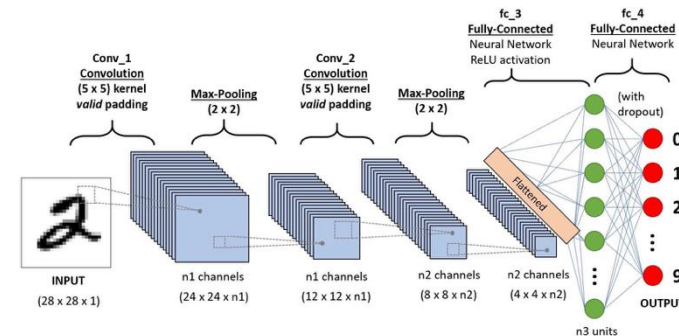
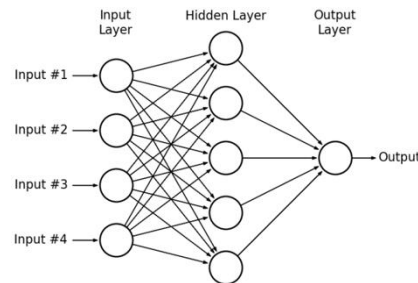
Convolutional Neural
Networks

2017

Transformer

Mathematical Model:

- Support Vector Machine
- Random Forest



EE2211
Introduction to Machine Learning

EE3703
Machine Learning with Applications

EE4212
Computer Vision

CS4248
Natural Language Processing



Content Table

1. Why use AI
2. History of Large Language Model (LLM)
3. Coding Agent -- VS Code Copilot
4. NUS AI Policy
5. Roadmap of Learning AI
6. Hands-on Activity



Hands on Activity

15 mins

DIY

Create a Python Notebook, write these down in the first cell.

```
years = [2018, 2019, 2020, 2021, 2022]
expenditure = [12.8, 12.7, 12.2, 12.9, 13.2]
```

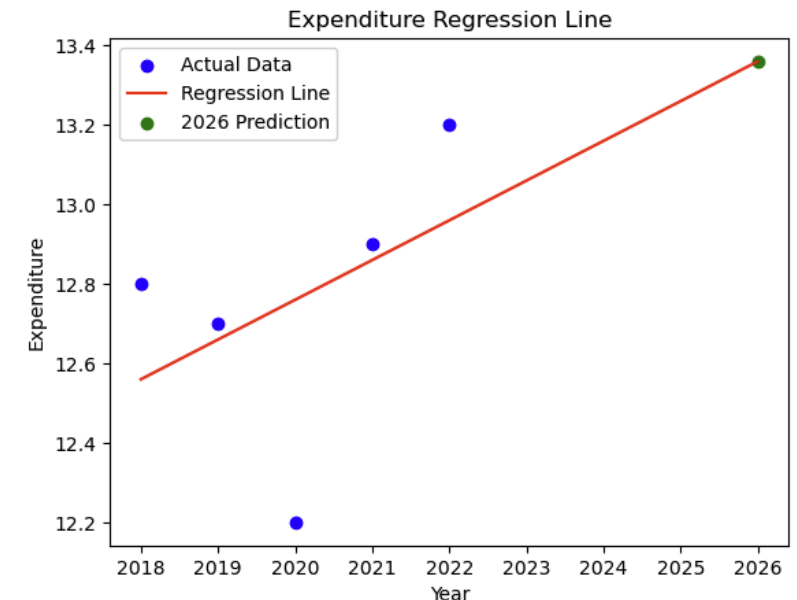
[4]

✓ 0.0s

Python

TODO:

1. Get a regression line from the past year data
2. Predict the value for 2026
3. Plot out the graph





Thank YOU